

ABM BUSINESS LOGIC BIBLE

§18 — EDGE CASES & EXCEPTION HANDLING

V2 Section 18 · Standalone Reconciliation · Two Complementary Catalogues, One Reference

§18 exists in two layers that were each built correctly but never assembled under one V2-numbered cover. V2's original text specifies four operational edge-case groups — signal, scoring, outreach, system — the kind of per-module exception a developer hits while implementing §3–§12. Our own Section 5 (§1.7/1.8 cross-cutting work) built a SEPARATE, complementary EDGE-01 through EDGE-16 catalogue: twelve named cross-cutting SCENARIOS (M&A, regulator action, sanctions, opt-out) that compose rules from across the whole Bible, gated by GOV's Autonomy Ladder. Both are real, both are necessary, and they answer different questions: V2's edge cases say 'what do I do when this specific module hits this specific weird input'; the EDGE catalogue says 'what does the WHOLE ENGINE do when reality itself gets strange.' §18 is where both live, clearly distinguished, in one place.

V2 edge case groups	4	EDGE scenarios (cross-cutting)	12
Rules (reconciled, both sources)	27	New rules (this pass)	8

WHY TWO CATALOGUES, NOT ONE

V2's edge cases are LOCAL: a signal mentions two accounts, a score hovers at a tier boundary, a contact leaves mid-sequence — each resolved by the module that owns the affected entity. The EDGE-01–16 catalogue is GLOBAL: an M&A, a sanctions hit, a source-poisoning attempt — each requires composing rules from MULTIPLE sections at once and almost always involves a GOV authority decision. Collapsing them into one undifferentiated list would lose this distinction and risk a developer treating a global scenario with a local fix, or vice versa.

18.0 V2 EDGE CASES — SIGNAL HANDLING

Rule ID	Specification
EDGE-SIG-001	Signal about an account NOT in the system: create a new account record (state WATCHING, tier COLD). Run the signal through normal processing. If the new account scores HOT immediately, follow the standard HOT flow. Source: V2 §18.1.
EDGE-SIG-002	Signal mentions MULTIPLE accounts (e.g. 'Bank A and Bank B announce joint digital initiative'): create one signal record per account; both get the same content but independent downstream processing. Source: V2 §18.1.
EDGE-SIG-003	Signal about a Decimal COMPETITOR winning a deal at the target account: NEGATIVE signal. Reduce relationship_warmth by 20 points on next score calculation (§4). account_state → DORMANT with reason 'Competitor won'. Cooldown: 180 days (double standard 90). Source: V2 §18.1. This is the rule already cited from §12 v1 CRM-PULL-04, §13 v2 ANL-MISS-02, §14 v1/v2 state tables, and §16 v1's timing matrix — §18 is its canonical specification; those are its consumers.
EDGE-SIG-004	Signal is in Arabic: AI translates to English for normalization; both the Arabic original AND the English translation are stored. confidence_score is reduced by 0.1 if machine-translated, until a human validates. Source: V2 §18.1.

18.1 V2 EDGE CASES — SCORING

Rule ID	Specification
EDGE-SCORE-001	Account has a perfect score (100) but zero contacts: tier capped at WARM regardless of score. The high score triggers priority enrichment (contact discovery), since outreach cannot begin without contacts. Source: V2 §18.2; restated as §5 v1 TIER-EDGE-01.
EDGE-SCORE-002	Account score fluctuates between 73–77 repeatedly (hovering at the HOT/WARM boundary): hysteresis band applies — once HOT, must drop to 70 (not 74) to demote; once WARM, must reach 78 (not 75) to promote. Source: V2 §18.2; this is the rule §5 v1/v2 fully specified (TIER-DEF-01, the asymmetric 78/70 band) — §18 cites it as the originating edge case, §5 as the authoritative mechanism.
EDGE-SCORE-003	Two accounts at the same banking group (e.g. a parent bank and its digital-banking subsidiary): treated as SEPARATE accounts with separate scores, linked via related_accounts. If one becomes HOT, flag the related account for human review on outreach coordination — never auto-promote the related account. Source: V2 §18.2; restated as §5 v1 TIER-EDGE-03.

18.2 V2 EDGE CASES — OUTREACH

Rule ID	Specification
EDGE-OUT-001	Contact leaves the company during an active sequence: if enrichment detects a role change, STOP all sequences for this contact; mark DISQUALIFIED with reason 'Left company'. Do NOT auto-create a new contact at their new company — that is governed separately by §3/§17 v2 SIG-MOBILITY/KSA-MOVE (a re-anchoring decision, not an automatic action). Source: V2 §18.3.
EDGE-OUT-002	Contact's email starts bouncing mid-sequence: after the first bounce, wait 48 hours and retry ONCE. If the second attempt also bounces, mark email INVALID and switch remaining touches to LinkedIn-only. If no LinkedIn exists, mark the contact UNREACHABLE. Source: V2 §18.3; consistent with §6 v1 ENR-EV-05/§9 v1 OUT-SEQ-04's bounce handling — this is the specific mid-sequence timing detail those rules assume.
EDGE-OUT-003	Two contacts at the same account reply on the same day: BOTH replies trigger handoff (§11). The handoff package shows BOTH replies. The account owner manages the coordination; the system does NOT send any auto-response to either contact. Source: V2 §18.3.
EDGE-OUT-004	Contact replies AFTER the sequence is complete (all touches sent, sequence closed): the reply is still classified per §10's sentiment rules, and handoff still triggers if positive. Sequence completion does not mean the account is abandoned. Source: V2 §18.3.

18.3 V2 EDGE CASES — SYSTEM

Rule ID	Specification
EDGE-SYS-001	AI model is down for an extended period (2+ hours): queue all AI tasks. Do NOT fall back to template-based messaging — template messages violate §7's 'reasoning before messaging' principle (P5). It is better to delay than to send ungrounded generic content. Source: V2 §18.4.
EDGE-SYS-002	Enrichment provider returns obviously wrong data (e.g. claims a major bank has 5 employees): flag the data SUSPICIOUS. Do NOT use it for scoring or outreach. Log for human review. Source: V2 §18.4; this is the data-layer instance of §6 v3's anti-fabrication discipline (ENR-BUDGET-01) — a provider hallucinating a fact is treated the same as the engine hallucinating one.
EDGE-SYS-003	Same person appears as a contact at two different accounts (e.g. a board member at two banks): allow the duplicate — they ARE two separate relationships. But coordinate outreach: never send to the same person from two different account sequences simultaneously. Minimum 14-day gap between touches across accounts. Source: V2 §18.4; this is the rule already cited in §16 v1's timing matrix (§16.3) — §18 is its canonical specification.

18.4 THE EDGE CATALOGUE (CROSS-CUTTING SCENARIOS) — RECONCILED FROM SECTION 5

Twelve named, GLOBAL scenarios — each composing rules from across the whole Bible, gated by GOV's Autonomy Ladder, each with a named detector, tier, and response. Transcribed here verbatim under their original IDs so §18 is complete as a standalone reference; nothing changed from their original specification.

ID	Scenario	Tier	Response (summary)
E1 (EDGE-03)	Target M&A	A4	Freeze sequences, invalidate opportunity graph, escalate before further outreach
E2 (EDGE-04)	Regulator action against entity	A4	Pause outreach immediately; resumption requires governance review
E3 (EDGE-05)	Executive death or scandal	A3	Halt messaging referencing exec; demote SIG-PATH anchors keyed on them
E4 (EDGE-06)	Vendor merger	—	Re-run displacement analysis; flag dependent dossier entries; no stale assertions
E5 (EDGE-07)	Sanctioned entity on path	A0	Hard-stop — send code path not invoked; never approval-eligible
E6 (EDGE-08)	Contradicting equally-trusted sources	A3	Withhold action; raise to Critical Unknown; never silently pick a side
E7 (EDGE-09)	Too-good-to-be-true (high score, low coverage)	A3	EPIS coverage cap forces approval regardless of tier
E8 (EDGE-10)	Departed exec joins a competitor	—	Re-classify relationship; preserve confidentiality; demote path to P5
E9 (EDGE-11)	Opt-out	A0	Add to do-not-contact; propagate across channels; never approval-eligible
E10 (EDGE-12)	Public crisis at target	—	Pause outreach; downgrade to relevance-only watching
E11 (EDGE-13)	Internal Decimal departure breaks a path	—	Emit path-break; halt in-flight sends keyed on that sender; re-resolve
E12 (EDGE-14)	Source poisoning / prompt injection	—	Quarantine source; reject for grounding; flag for OBS review

Rule ID	Specification
EDGE-01	Every EDGE scenario has a named detector (signal + EPIS condition), a named tier (A0–A5), and a named response composed of existing rules from across the Bible.
EDGE-02	An active EDGE scenario at an account preempts the normal EXEC sequence at the earliest stage requiring escalation; later stages may not execute until resolved.
EDGE-15	Every EDGE response is recorded as a Failure Pattern Record (§4 v2 FAIL-LEARN-01 / §15 v2's generalized Repeat-Failure mechanism) when it implies a model gap — a scenario the engine should have detected earlier.
EDGE-16	The EDGE catalogue is reviewed at every section update; new scenarios encountered in operation are added with detector + tier + response before they recur — the same discipline §15 v2 FAIL-INVEST-01's Resilience Opportunity Register applies to operational failures, applied here to edge scenarios.

18.5 RECONCILING THE TWO CATALOGUES

Eight rules close the gap between V2's local edge cases and the EDGE catalogue's global scenarios — both confirming where they're genuinely distinct and flagging the one place they could be confused.

Rule ID	Specification
EDGE-RECON-01	V2's edge cases (§18.0–18.3) are LOCAL exceptions, resolved within the owning module's existing rules, never requiring GOV escalation or cross-section composition. The EDGE-01–16 catalogue (§18.4) is GLOBAL, composes rules from multiple sections, and is almost always GOV-gated. A developer encountering an exception should first ask which category it is — this rule is the explicit test.
EDGE-RECON-02	EDGE-SIG-003 (V2, competitor-win) and E4/EDGE-06 (vendor merger, cross-cutting) can both fire on the same underlying market event (a competitor displacing Decimal at an account that ALSO involves a vendor change) — they are not mutually exclusive and may both apply: EDGE-SIG-003 governs the local relationship_warmth/cooldown mechanics; EDGE-06 governs the global displacement-analysis re-run. Neither supersedes the other.
EDGE-RECON-03	A LOCAL edge case that recurs with unusual frequency or severity (per §15 v2 FAIL-REPEAT-02's escalation discipline) is itself grounds to propose promoting it to a new GLOBAL EDGE-catalogue entry — this is the formal promotion path from §18.0–18.3 into §18.4, submitted as an A4 governance proposal (GOV-MOD-01) like any other architectural change, never informally.
EDGE-RECON-04	Both catalogues share ONE governing principle, restated here as §18's binding test: an exception-handling rule that does not specify what the system should DO (not merely what went wrong) is incomplete — every entry in §18.0–18.4 names a concrete action, never only a description of the failure mode.

5-WHYS — WHY §18 NEEDS BOTH CATALOGUES IN ONE PLACE

Why #1: Why keep V2's local edge cases and the EDGE-01–16 global catalogue as two separate lists rather than merging them into one numbered sequence?

Because merging them would erase the exact distinction that makes each useful: a developer fixing a bounced-email mid-sequence bug (EDGE-OUT-002) needs a local, module-level answer fast; a developer building the GOV escalation pathway for a sanctioned-entity hit (E5) needs to understand cross-section composition and Autonomy Tiers. Numbering them into one undifferentiated sequence would invite treating a global scenario with a local fix — exactly the failure mode §18.5 EDGE-RECON-01 exists to prevent.

Why #2: Why does EDGE-SIG-003 (competitor-win) get its full specification here rather than wherever it happened to be needed first?

Because it was already being CITED by name from §12, §13 v2, §14, and §16 before this document existed — each citation assumed a canonical source that didn't actually have its own home. This is the same structural gap §15 v1 found and fixed for the FAIL-* rules: a rule cited from five places needs to live SOMEWHERE as its primary specification, or the citations are pointing at nothing.

Why #3: Why must EDGE-SYS-001 explicitly forbid falling back to template messaging during an extended AI outage, rather than treating any output as better than no output?

Because §7's Principle 5 (reasoning before messaging) is not a style preference — a template message sent without the grounded reasoning chain Agent 1–5 produce is exactly the ungrounded, hallucination-risk output the entire §7/§8 architecture exists to prevent. Outage-induced pressure to 'just send something' is precisely when this discipline is most tempting to abandon and most important to hold.

Why #4: Why does a recurring LOCAL edge case get a formal promotion path into the GLOBAL catalogue rather than just being fixed repeatedly where it occurs?

Because §15 v2's Repeat Failure theory already established that a recurring issue is evidence the ORIGINAL fix was insufficient, not evidence the fix needs to be reapplied identically again. A local edge case recurring with unusual severity may indicate it was never truly local — it may be a symptom of a broader pattern that deserves the cross-section composition and GOV visibility the global catalogue provides.

Why #5: Why must every §18 entry specify a concrete action rather than just documenting the failure mode?

Because a catalogue of named-but-unresolved exceptions is a list of known unknowns, not a specification — and this Bible has consistently distinguished between an OPEN-DESIGN-QUESTION (a genuinely undecided choice, flagged honestly) and an edge case (a scenario with a knowable, decidable correct response). §18's entries are all the latter; conflating the two categories would let unresolved design questions hide inside what should be a complete, actionable reference.

Root cause: edge-case handling needs both a local, module-level catalogue for fast operational fixes and a global, GOV-gated catalogue for cross-cutting scenarios, each cited rule needs one canonical home rather than being assumed from multiple citing locations, discipline-preserving rules (like never templating during an AI outage) must hold especially firm under the exact pressure that tempts abandoning them, and every entry must specify action, not just failure description.

§18 RULE ROLL-UP & CROSS-LINKS

Group	Rules	Key cross-links
V2 Signal edge cases	EDGE-SIG-001..004	§12 v1 CRM-PULL-04, §13 v2 ANL-MISS-02, §14/§16 (cited from)
V2 Scoring edge cases	EDGE-SCORE-001..003	§5 v1 TIER-EDGE-01/03, §5 hysteresis (TIER-DEF-01)
V2 Outreach edge cases	EDGE-OUT-001..004	§3/§17 v2 SIG-MOBILITY, §6 v1 ENR-EV-05, §9 v1 OUT-SEQ-04, §11
V2 System edge cases	EDGE-SYS-001..003	§7 Principle 5, §6 v3 ENR-BUDGET-01, §16 v1 timing matrix
EDGE catalogue (global)	EDGE-01..16	GOV Autonomy Ladder, §4 v2 FAIL-LEARN-01, §15 v2 FAIL-REPEAT/INVEST
Reconciliation	EDGE-RECON-01..04	Promotion path local→global, GOV-MOD-01

§18 COMPLETE — WHAT IT ESTABLISHED

§18 now exists as its own standalone document carrying BOTH of the Bible's edge-case layers, clearly distinguished: V2's four local catalogues (signal, scoring, outreach, system — 15 rules, each resolved within its owning module) and the EDGE-01–16 global catalogue (12 cross-cutting scenarios, each GOV-gated and composed from multiple sections). Several V2 rules that had been cited from five different sections without ever having a canonical home (EDGE-SIG-003 most prominently) now have one. A formal promotion path lets a recurring local edge case become a global catalogue entry through proper governance, and one shared test — every entry specifies an action, not just a failure description — binds both catalogues together as one coherent discipline. 8 new reconciliation rules.

With §18 built as its own document, the same findability gap §15 closed for failure-handling is now closed for edge-case handling — both were architecturally correct and complete; neither was easy to find under its own V2 section number until now.

PLACEMENT MAP UPDATE

§18 status: confirmed Complete, now WITH a standalone artifact (previously Complete-but-distributed across §5/§12/§13/§14/§16 citations plus the original Section 5 EDGE catalogue). No open questions — this was reconciliation, not new design.

ON THE HORIZON

With §1–§18 now each having a standalone, findable artifact, the only remaining V2-numbered item is §19 Glossary of Business Terms — merging V2's original term list with every new term this Bible has coined since (RCM Stamp, Effective Path, Reality Convergence, Sponsor vs Champion, and dozens more).